

HARSHITA KUMARI

Computer Science Student

tiwaryharshita18@gmail.com | 8809019283 | Jamshedpur, Jharkhand

LinkedIn: in/harshita-kumari | GitHub: github.com/harshita

EDUCATION

B.Tech in Computer Science Engineering

2023 – 2027

Birla Institute of Technology, Mesra, Jharkhand

CGPA: 8.34/10

EXPERIENCE

Data Science Intern

May 2025 – July 2025

Tata Motors, Jamshedpur, Jharkhand

- Applied Machine Learning techniques and Python-based data processing to analyze and preprocess image datasets, supporting downstream model training and evaluation workflows.
- Implemented a U-Net CNN in PyTorch for pixel-level semantic segmentation; gained hands-on experience with dataset handling, training pipelines, and model performance evaluation using IoU metrics.

PROJECTS

Road Extraction Using Satellite Images (U-Net Based)

- Implemented a U-Net Convolutional Neural Network (CNN) in PyTorch to perform pixel-level semantic segmentation for road extraction from high-resolution satellite images using binary masks.
- Built a data preprocessing and evaluation pipeline involving image normalization, mask alignment, and model performance analysis using Intersection over Union (IoU), along with visualization of predicted vs. ground-truth masks for qualitative and quantitative assessment.

RAG-Based AI Teaching Assistant

- Built a Retrieval-Augmented Generation (RAG) pipeline to transform lecture videos into searchable knowledge using audio transcription, timestamped text chunking, and metadata processing with Whisper.
- Implemented semantic embedding-based retrieval to fetch top-K relevant chunks and supply them to an LLM via context-aware prompting, improving answer accuracy and reducing hallucinations.

ACHIEVEMENTS

- Semi-Finalist, **Myntra HackRamp** — ranked among the top students out of 1.5 lakh+ participants nationwide.
- 2nd place** at Data Science Summit'24 Hackathon out of 150+ teams.
- Solved **400+ problems** on LeetCode, demonstrating strong Data Structures & Algorithms skills.

SKILLS

Languages	Java, Python, JavaScript, C
Web Development	React.js, HTML5, Tailwind CSS, Node.js, REST APIs
ML & AI	PyTorch, U-Net, Semantic Segmentation, RAG, Embeddings, LLMs, IoU
Data & Visualization	Pandas, NumPy, Matplotlib, OpenCV
Databases	MongoDB, SQL

CERTIFICATIONS

- The Ultimate Job Ready Data Science Course — CodeWithHarry (2025)
- Complete 2025 Python Bootcamp: Learn Python from Scratch — CodeWithHarry (2025)

POSITIONS OF RESPONSIBILITY

- General Secretary, Robotics Society** — Led society operations and spearheaded technical events and workshops.
- Management Lead, Akshar Literary Society** — Owned planning and execution of campus literary events.
- Lead, Drishti (NGO)** — Coordinated community outreach and volunteer-led social initiatives